AI-Based Medical Diagnosis System

1. Summary:

The proposed AI model acts as a virtual medical assistant, diagnosing diseases, predicting health issues, and recommending solutions based on user symptoms and follow-up questions. It mimics a doctor's diagnostic approach, making it user-friendly for laymen. With high accuracy, it can provide quick, accessible healthcare insights, especially in remote or underserved areas. Proper implementation, data privacy, and integration with medical professionals are key to its success. This innovation has the potential to revolutionize healthcare by improving efficiency, reducing costs, and empowering users with timely medical guidance.

2. Introduction:

Technology and healthcare are coming together to improve disease prevention and early detection. With people living longer, chronic diseases are becoming more common, making health screenings more important. Using big data and AI, modern systems can predict diseases more accurately, helping doctors make better decisions and improving healthcare efficiency.

3. Related Work:

Zhang, Fang, Zhen Zhang, and Hui Xiao. ***"Research on medical big data analysis and disease prediction method based on artificial intelligence."*** *Computational and Mathematical Methods in Medicine 2022.*

4. Methodology Steps:

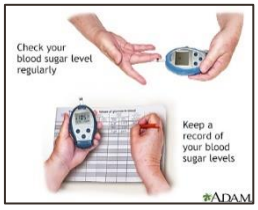
- **Data Collection:** Gathered 21,140 health records with 27 key features.
- **Data Preprocessing:** Cleaned, normalized, and selected relevant features.
- **Data Augmentation:** Generated synthetic data to enhance model learning.
- **Model Development:** Used Random Forest and CatBoost algorithms for prediction.
- **Training & Testing:** Split data into learning and evaluation sets.
- **Performance Metrics:** Evaluated using Accuracy, ROC Curve, Precision, and Recall.
- **Business Application:** Designed for integration into commercial healthcare platforms.

5. Dataset:

The dataset consists of 21,140 health check-up records**,** including diagnostic test results for chronic diseases like diabetes and gout**.**

Data Processing

Collection of Healthcare



Building Healthcare DBMS

	Factor 1	Factor2	Factor 3
Patien 1	330	80	89
Patient 2	110	75	95
Patient 3	97	85	75

Data Design & Labelling

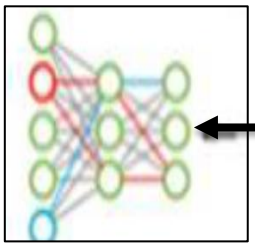
	Factor 1	Factor2	Factor 3
Patient1	330	80	89
Patient2	110	75	95
Patient3	97	85	75

Establishment of AI model

Weighted Value Inversion System Design

Factor 1 0.1

Factor 2 0.6



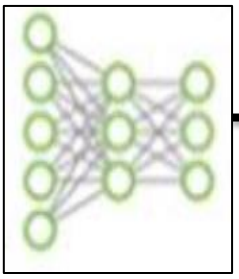
Disease 1

Weiaht ← Disease

AI Learning System

Factor 1 0.1

Factor 2 0.6



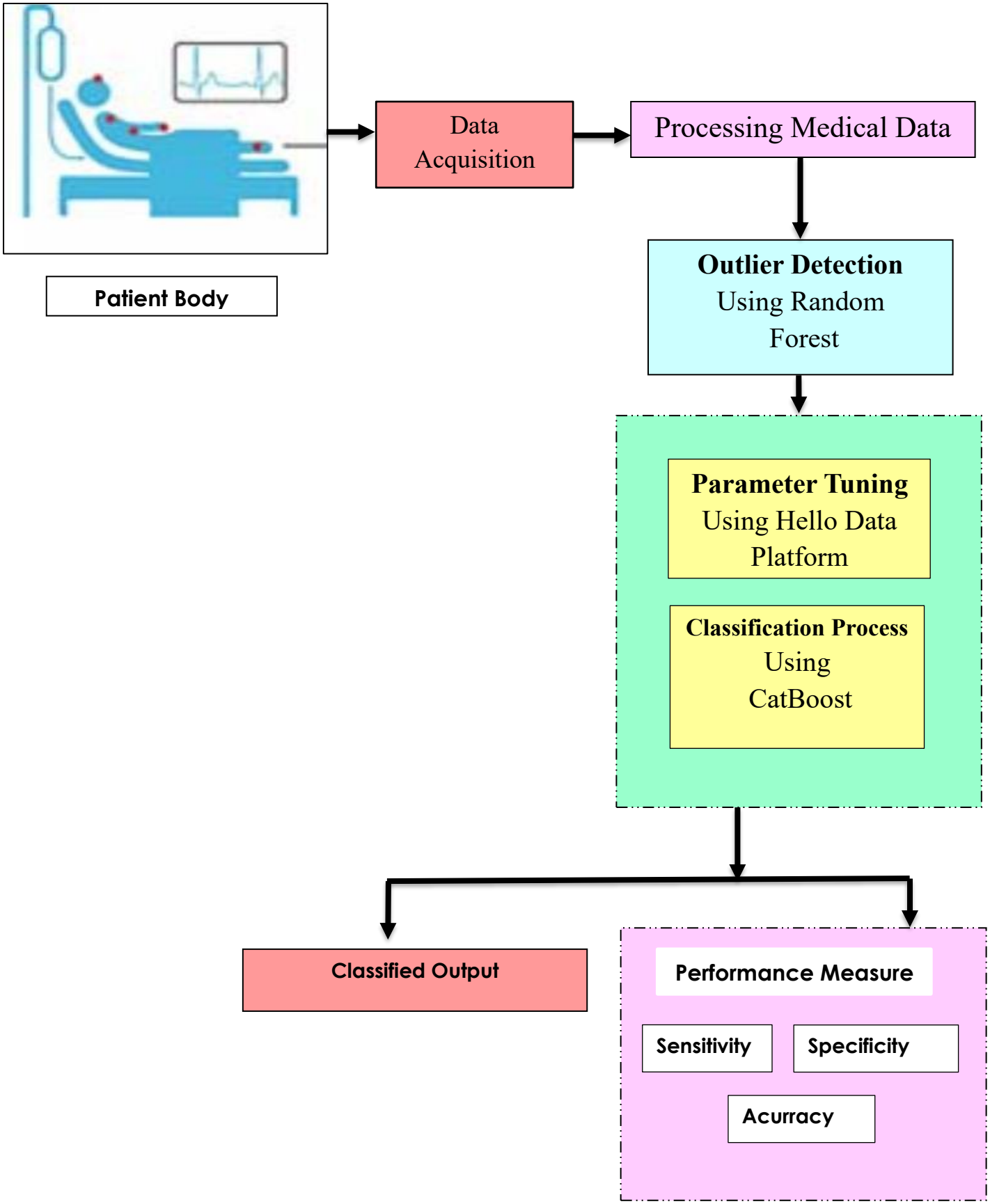
Results

Patient1 → Disease1

Business Application

	Factor 1	Factor 2 ----------Factor 23				Check
Patient 3	95	85	----------	75	Disease 3	☑
Patient 2	100	80	----------	70	Normal	☐
Patient 1	110	75	----------	65	Normal	☐

Research Procedure


Patient Body
Data Acquisition
Processing Medical Data
Outlier Detection Using Random Forest
Parameter Tuning Using Hello Data Platform
Classification Process Using CatBoost
Classified Output
Performance Measure
Sensitivity
Specificity
Acurracy

6. Performance Measures:

The AI model's performance was evaluated using the following metrics:

- ➔ **Accuracy:**
  Measures how correctly the model predicts diseases (Diabetes: 91%, Gout: 98%).
- ➔ **ROC Curve & AUC (Area Under Curve):**
  Evaluates classification performance (Diabetes: 0.92, Gout: 0.99).
- ➔ **Confusion Matrix:**
  Assesses True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).
- ➔ **Precision:**
  The proportion of correct positive predictions (Diabetes: 0.88, Gout: 0.97).
- ➔ **Recall:**
  The ability to identify actual cases (Diabetes: 0.82, Gout: 0.63).
- ➔ **False Positive Rate (FP Rate):**
  Measures incorrect disease predictions (Diabetes: 12%, Gout: 2.4%).
- ➔ **True Positive Rate (TP Rate):**
  Correctly identified disease cases (Diabetes: 88%, Gout: 98%).

These metrics validate the model's effectiveness in **disease diagnosis and prediction**.

7. Reference:

Park, Jae Young, and Jai-Woo Oh. ***"****Original Research Article Study on prediction and diagnosis AI model of frequent chronic diseases based on health checkup big data.****" Journal of Autonomous Intelligence 7.3(2024),*** front-sci.